Exercise 1: One-Round Game (2 Human Players)

- **How it plays:** Two human players simultaneously choose to Cooperate (C) or Defect (D) exactly once.
- **Behavior & Dynamics:**
    - **Strict Dominance:** Regardless of what the other player chooses, defection yields a higher personal score:
        - If Player B cooperates, defecting gives 5 instead of 3.
        - If Player B defects, defecting gives 1 instead of 0.
    - **Nash Equilibrium:** Rational self-interest drives both players to defect, resulting in (D, D) with a payout of $(1, 1)$.
    - **Social Dilemma:** Even though mutual cooperation (C, C) yields a higher combined and individual payoff $(3, 3)$, fear of betrayal (getting 0) or greed (aiming for 5) prevents cooperation in an isolated round.

Exercise 2: N-Rounds Game (2 Human Players)

- **How it plays:** The same two human players face each other across a fixed number of rounds (n), observing previous moves as the game unfolds.
- **Behavior & Dynamics:**
    - **Reputation & Trust:** Because actions have long-term consequences, players can signal cooperation early on to achieve mutual $(3, 3)$ payoffs.
    - **Retaliation & Distrust:** A single defection usually triggers immediate retaliation in subsequent rounds, causing the game to collapse into a cycle of mutual defection $(1, 1)$.
    - **The "End-Game" Effect:** If both players know exactly which round is the last, there is no threat of future punishment on round n. As a result, both have an incentive to defect on the final round. Through backward induction, this distrust can cascade backward into earlier rounds.

Exercise 3: Human vs. Computer (Strategy Testing)

- **How it plays:** A human player tests their decision-making against an automated computer following one of four distinct strategies:
- **Behavior by Strategy:**
    - **Strategy 1: Always Cooperate**
        - *Behavior:* Always chooses C regardless of what you do.
        - *How to exploit:* The human can play D every round to receive the maximum possible score of 5 per round, while the bot receives 0.
    - **Strategy 2: Always Defect**
        - *Behavior:* Always chooses D regardless of what you do.
        - *Optimal response:* The human must also play D every round to receive 1 point. Cooperating yields the sucker's payoff of 0.
    - **Strategy 3: Random**
        - *Behavior:* Flips a coin between C and D.

        - *Optimal response:* The computer cannot be trained or reasoned with. Playing D yields an expected value of 3 per round ($0.5 \times 5 + 0.5 \times 1$), compared to 1.5 for playing C ($0.5 \times 3 + 0.5 \times 0$). Constant defection remains mathematically optimal.
    - **Strategy 4: Tit-for-Tat**
        - *Behavior:* Starts with C, then mimics your previous round's move.
        - *Dynamics:* Encourages mutual cooperation. If you play C, it rewards you with C (3 points each). If you defect, it punishes you in the very next turn with D. It is forgiving: as soon as you switch back to C, it resumes cooperating.

Exercise 4: Computer vs. Computer (Random Strategy Matchups)

- **How it plays:** Two computer agents are randomly assigned one of the four strategies above and play against each other.
- **Pairing Behaviors:**
    - **Tit-for-Tat vs. Tit-for-Tat:** Both start with C and continue playing C forever, achieving optimal mutual payoffs of 3 points per round.
    - **Tit-for-Tat vs. Always Defect:** Tit-for-Tat cooperates on round 1 and loses (0 vs 5). From round 2 onward, Tit-for-Tat defects in response, locking the match into $(1, 1)$ payoffs for all remaining rounds.
    - **Always Cooperate vs. Always Defect:** Complete exploitation—Always Defect earns 5 every round, while Always Cooperate earns 0.
    - **Random vs. Any Strategy:** Random disrupts adaptive strategies like Tit-for-Tat by introducing chaos and triggering unintended cycles of defection.

Exercise 5: Single-Elimination Knockout Tournament

- **How it plays:** An even number of automated players are paired off in a bracket; each pair plays a series of rounds, and only the player with the higher total score advances to the next stage until one champion remains.
- **Behavior & Dynamics:**
    - **Exploitation Advantage:** In head-to-head elimination matches, an **Always Defect** bot will beat an **Always Cooperate** bot (5 to 0 per round) and move forward.
    - **The Head-to-Head Limitation of Tit-for-Tat:** Tit-for-Tat can never score *more* points than its opponent in a single match (it can either tie them or lose by at most one defection). In a knockout format where advancing depends purely on having a higher score than your direct opponent, Always Defect often advances over cooperative bots.
    - **Contrasting Tournament Formats:** In Axelrod's famous round-robin tournament (where total points across all matches determine the winner), Tit-for-Tat wins because mutual cooperation accumulates high point totals against other cooperators. In a sudden-death elimination bracket, aggressive/defecting strategies often survive longer against naive cooperators